Characteristic-free constructive set-multilinearization preserving low product depth

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Allowed standard prerequisites (editorial).

- Binet-Minc identity for rectangular permanents. Explicitly quoted author Theorem13 citing Minc[17]; full identity retained.
- Bell-number asymptotics and elementary partition counting. Author Fact15 citing Flajolet,Sedgewick[6], Definition14 and Lemma16; exact statements retained.

Author circuit and polynomial conventions

Algebraic complexity asks for the best methods to compute multivariate polynomials $f(x_1, \ldots, x_n)$ from the primitive operations of addition and multiplication, using constants from the underlying field $\mathbb{F}$ for free. These computations are organized as an algebraic circuit, a directed acyclic graph of these primitive operations from the input variables $x_1, \ldots, x_n$. The complexity of this object is primarily governed by its size, which is the number of operations in the circuit. Another complexity measure is the product depth, which is the maximum number of product operations on any input to output path. The iterated matrix multiplication polynomial $\mathrm{IMM}_{n,d} = (X_1 \cdots X_d)_{1,1}$ where X_i is an $n \times n$ symbolic matrix. F_k is the k-th Fibonacci number (so $F_0 = 0$, $F_1 = F_2 = 1$, $F_3 = 2$, etc.).

Definition 7. Let the variables $\bar{x}$ be partitioned into $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$. A monomial is set-multilinear (with respect to the partition $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$) if it can be written has $\prod_{i=1}^{d} (\bar{x}_i)_{j_i}$ for some $j_1, \ldots, j_d$. Define the set-multilinear projection (with respect to the partition $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$) to be the linear map $\pi_{\mathrm{sm}} : \mathbb{F}[\bar{x}] \to \mathbb{F}[\bar{x}]$ which is identity on set-multilinear monomials, and zero on all other monomials.

Definition 11. Let X be an $n \times m$ symbolic matrix of variables, with $n \leq m$, where $X_{i,j} = x_{i,j}$ are distinct variables. Define the (rectangular) permanent

$\mathrm{perm}(X) \in \mathbb{Z}[(x_{i,j})_{i\in[n],j\in[m]}]$ by

$$\mathrm{perm}_{n\times m}(X) = \sum_{\sigma:[n]\hookrightarrow[m]} x_{1,\sigma(1)} \cdots x_{n,\sigma(n)},$$

that is, the sum runs over all injective maps σ from $[n]$ to $[m]$. Note in particular that $\mathrm{perm}_{n\times m}$ is a set-multilinear polynomial when we partition the matrix into its rows. It is sometimes helpful to view the rectangular permanent as a sum of square permanents.

Lemma 12. Let X be an $n \times m$ symbolic matrix of variables, with $n \leq m$, where $X_{i,j} = x_{i,j}$ are distinct variables. Then,

$$\mathrm{perm}_{n\times m}(X) = \sum_{S\in\binom{[m]}{n}} \mathrm{perm}_{n\times n}(X|_{[n]\times S}).$$

The above lemma shows the rectangular permanent is computable by $\mathrm{poly}(m^n)$ size algebraic circuits. However, the following identity gives a non-trivially better algorithm.

Theorem 13 (Binet-Minc Identity [17]). Let X be an $n \times m$ symbolic matrix of variables, with $n \leq m$, then the permanent can be computed by

$$\mathrm{perm}_{n\times m}(X) = \sum_{\mathcal{F}\in\mathcal{P}_n} (-1)^{n-|\mathcal{F}|} \prod_{S\in\mathcal{F}} (|S|-1)! \sum_{j=1}^{m} \prod_{i\in S} x_{i,j}$$

where $\mathcal{P}_n$ is the set of all partitions of $[n]$ into (non-empty) sets, and $|\mathcal{F}|$ is the number of parts in the partition $\mathcal{F}$ of $[n]$. To understand the complexity of this expression it is helpful to have the following definition.

Definition 14. Define the n-th Bell number B_n to be the number of ways to partition $[n]$ into (non-empty) sets. We will use the following asymptotic estimate of Bell number size.

Fact 15 ([6]). $B_n = \Theta(\frac{n}{\ln n})^n$. The following lemma is easy to prove from the definition of Bell numbers.

Lemma 16. $B_n \cdot B_m \leq B_{n+m}$. The Binet-Minc identity immediately implies the following algebraic circuit for the permanent.

Corollary 17. The rectangular permanent $\mathrm{perm}_{n\times m}$ has a $\Sigma^{B_n}\Pi^n\Sigma^m\Pi^n$ formula of size $\mathrm{poly}(m, B_n)$, where the super-scripts are upper bounds on the respective fan-ins of the formula. Further, this formula is set-multilinear (and hence homogeneous) with respect to the partition of the $n \times m$ variables into rows.

Definition 20. Let X be an $n \times m$ symbolic matrix of variables, with $n \leq m$, where $X_{i,j} = x_{i,j}$ are distinct variables. Let $k \leq n$. Define the k-surjective (rectangular) permanent $\mathrm{perm}_{n\times m;k}(X) \in \mathbb{Z}[(x_{i,j})_{i\in[n],j\in[m]}]$ by

$$\mathrm{perm}_{n\times m;k}(X) = \sum_{\sigma:[n]\hookrightarrow[m],\mathrm{im}(\sigma)\supseteq[k]} x_{1,\sigma(1)}\cdots x_{n,\sigma(n)},$$

that is, the sum runs over all injective maps σ from $[n]$ to $[m]$ that contain $[k]$ in their image. Note in particular that $\mathrm{perm}_{n\times m;k}$ is a set-multilinear polynomial when we partition the matrix into its rows. We can compute the surjective permanent by standard permanents.

Definition 23. Let the variables $\bar{x}$ be partitioned into $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$. Let $S \subseteq [d]$. A monomial is S-set-multilinear if it can be written as $\prod_{i\in S}(\bar{x}_i)_{j_i}$ for some $(j_i)_{i\in S}$. Define the S-set-multilinear projection to be the linear map $\pi_{\mathrm{sm},S}$ which is identity on S-set-multilinear monomials, and zero on all other monomials. The set of S-set-multilinear monomials for some $S \subseteq [d]$ are called at most set-multilinear monomials. Define the non-set-multilinear projection to be the linear map $\pi_{\neg\mathrm{sm}}$ which is zero on at-most set-multilinear monomials, and identity on all other monomials. In particular, for $S = [d]$, $\pi_{\mathrm{sm}}(f)$ and $\pi_{\mathrm{sm},S}(f)$ are the same. For $S = \varnothing$, $\pi_{\mathrm{sm},S}(f)$ is the constant part of f, $\pi_{\mathrm{sm},\varnothing}(f) = f(0)$. More generally, we can decompose f into its set-multilinear parts as follows.

Corollary 27. Let $\mathbb{F}$ be any field. Let the variables $\bar{x}$ be partitioned into $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$. Suppose $f \in \mathbb{F}[\bar{x}]$ can be computed by a size s product-depth Δ algebraic circuit. Then the set-multilinear projection $\pi_{\mathrm{sm}}(f) \in \mathbb{F}[\bar{x}]$ can be computed by a size $\mathrm{poly}(s, \Theta(\frac{d}{\ln d})^d)$-size product-depth 2Δ circuit.

Complete author-local construction supporting Corollary 27

Lemma 21. Let X be an $n \times m$ symbolic matrix of variables, with $n \leq m$, where $X_{i,j} = x_{i,j}$ are distinct variables. Let $k \leq n$. Then, the k-surjective permanent can be written as

$$\mathrm{perm}_{n\times m;k}(X) = \sum_{S\in\binom{[n]}{k}} \mathrm{perm}_{k\times k}(X|_{S\times[k]})\cdot\mathrm{perm}_{(n-k)\times(m-k)}(X|_{([n]\setminus S)\times([m]\setminus[k])}).$$

Further, this expression is set-multilinear with respect to partitioning X by its rows.

Proof. This follows from the observation that each map $\sigma : [n] \hookrightarrow [m]$ with $\mathrm{im}(\sigma) \supseteq [k]$ uniquely decomposes into a bijection $\tau : S \leftrightarrow [k]$ for some $S \in \binom{[n]}{k}$, and an injection $\nu : ([n] \setminus S) \hookrightarrow ([m] \setminus [k])$. The claim about set-multilinearity then follows from noting that the multiplication $\mathrm{perm}_{k\times k}(X|_{S\times[d]})\cdot\mathrm{perm}_{(n-k)\times m-k}(X|_{([n]\setminus S)\times([m]\setminus[k])})$ is a product of two polynomials who only use disjoint rows of X. □

We now analyze the complexity of computing the surjective permanent, by reduction to standard permanents, and then applying the Binet-Minc identity.

Corollary 22. Let $k \leq n \leq m$. The $n \times m$ k-surjective permanent has a $\mathrm{poly}(m, \Theta(\frac{n}{\ln n})^n)$-size depth-4 formula that is set-multilinear with respect to rows.

Proof. Via the above lemma, and the formula complexity of the Binet-Minc identity,

$$\mathrm{perm}_{n\times m;k}(X) = \sum_{S\in\binom{[n]}{k}} \underbrace{\mathrm{perm}_{k\times k}(X|_{S\times[k]})}_{\Sigma^{B_k}\Pi^k\Sigma^k\Pi^k} \cdot \underbrace{\mathrm{perm}_{(n-k)\times(m-k)}(X|_{([n]\setminus S)\times([m]\setminus[k])})}_{\Sigma^{B_{n-k}}\Pi^{n-k}\Sigma^{m-k}\Pi^{n-k}},$$

distributing the multiplication past the addition,

$$= \underbrace{\sum_{\binom{n}{k}} \sum^{B_k\cdot B_{n-k}}}_{\Sigma^{\binom{n}{k}B_kB_{n-k}}} \underbrace{\left(\prod^k \sum^k \prod^k\right) \cdot \left(\prod^{n-k} \sum^{m-k} \prod^{n-k}\right)}_{\Pi^n\Sigma^m\Pi^n}$$

from which the size bound follows by noting that $\binom{n}{k}B_kB_{n-k} \leq 2^nB_n \leq \Theta(\frac{n}{\ln n})^n$ via Lemma 16 and Fact 15. The set-multilinearity of this formula follows from the fact that the decomposition used here from the above lemma is set-multilinear, that Binet-Minc is set-multilinear, and that our use of the distributive law preserves set-multilinearity. □

We now proceed to give a non-trivial set-multilinearization for low-depth algebraic circuits. To do so, it will be helpful to have more notation for extracting various set-multilinear components of polynomials.

Lemma 24. Let the variables $\bar{x}$ be partitioned into $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$. Then for any polynomial f,

$$f = \pi_{\neg\mathrm{sm}}(f) + f(0) + \sum_{\varnothing\neq S\subseteq[d]} \pi_{\mathrm{sm},S}(f).$$

It immediately follows that we can simulate an addition (or even a linear combination) of polynomials by instead adding the constituent set-multilinear parts.

Lemma 25. Let f be any field. Let the variables $\bar{x}$ be partitioned into $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$. Let $f_1, \ldots, f_m$ be polynomials, with $f = \alpha_1 f_1 + \cdots + \alpha_m f_m$ for $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$. Let $S \subseteq [d]$. Then $\pi_{\mathrm{sm},S}(f)$ can be computed by a depth-1 $\mathrm{poly}(m, 2^d)$-size set-multilinear Σ-circuit given $\{\pi_{\mathrm{sm},S}(f_j)\}_{S\subseteq[d],j\in[m]}$ as inputs. Less trivially is the simulation of a multiplication, for which we use the formula for the surjective permanent (Corollary 22).

Lemma 26. Let the variables $\bar{x}$ be partitioned into $\bar{x} = \bar{x}_1 \sqcup \cdots \sqcup \bar{x}_d$. Let $f_1, \ldots, f_m$ be polynomials, with $f = f_1 \cdot \cdots \cdot f_m$. Let $S \subseteq [d]$ be of size ℓ. Then $\pi_{\mathrm{sm},S}(f)$ can be computed by a depth-4 $\mathrm{poly}(m, \Theta(\frac{\ell}{\ln \ell})^\ell, 2^d)$-size set-multilinear $\Sigma\Pi\Sigma\Pi$-circuit given $\{\pi_{\mathrm{sm},S}(f_j)\}_{S\subseteq[d],j\in[m]}$ as inputs.

Proof. The number of inputs to the circuit is $m2^d$. It remains to bound the number of gates by $\mathrm{poly}(m, \Theta(\frac{\ell}{\ln \ell})^\ell)$. Rearrange the f_i as needed so that $f_1(0) =$

$\cdots = f_k(0) = 0$ and $f_{k+1}(0), \ldots, f_m(0) \neq 0$, for some $k \leq m$. Thus, we can normalize the computation via

$$f = \prod_{i>k}^{m} f_i(0) \cdot \prod_{i \in [k]} f_i \cdot \prod_{i>k}^{m} \frac{f_i}{f_i(0)},$$

and thus define

$$g_i = \begin{cases} f_i & i \leq k \\ f_i / f_i(0) & i > k \end{cases},$$

and $g = \prod_i g_i$. It then follows that $f = \prod_{i>k}^{m} f_i(0) \cdot g$, $g_{k+1}(0) = \cdots = g_m(0) = 1$, and $\pi_{\mathrm{sm},S}(f) = \prod_{i>k}^{m} f_i(0) \cdot \pi_{\mathrm{sm},S}(g)$. As $\prod_{i>k}^{m} f_i(0)$ is a non-zero constant, it suffices to prove the claim for $\pi_{\mathrm{sm},S}(g)$. Now write g as

$$g = \prod_{i=1}^{k} \left(\pi_{\neg \mathrm{sm}}(g_i) + \sum_{\varnothing \neq S_i \subseteq [d]} \pi_{\mathrm{sm},S_i}(g_i) \right) \cdot \prod_{i>k}^{m} \left(\pi_{\neg \mathrm{sm}}(g_i) + 1 + \sum_{\varnothing \neq S_i \subseteq [d]} \pi_{\mathrm{sm},S_i}(g_i) \right).$$

In expanding the above product, the only at-most set-multilinear terms are of the form

$$\prod_j \pi_{\mathrm{sm},S_{i_j}}(g_{i_j}),$$

where the $S_{i_j} \subseteq [d]$ are disjoint (as otherwise we create non-multilinear terms), and the indices $\{i_j\}_j$ are distinct (as we take exactly one term from each g_i [possibly the term 1]). Further, as $g_i(0) = 0$ for $i \leq k$, we must have $\{i_j\}_j \supseteq [k]$. Hence, by collecting like terms, we see that

$$\pi_{\mathrm{sm},S}(g) = \sum_{\mathcal{F} \in \mathcal{P}_S} \sum_{\sigma : \mathcal{F} \hookrightarrow [m]; \mathrm{im}(\sigma) \supseteq [k]} \prod_{T \in \mathcal{F}} \pi_{\mathrm{sm},T}(g_{\sigma(T)})$$

where $\mathcal{P}_S$ is the collection of set partitions of S, which we can rewrite in terms of k-surjective permanents as

$$= \sum_{\mathcal{F} \in \mathcal{P}_S} \mathrm{perm}_{|\mathcal{F}| \times m; k}(A_{\mathcal{F}}),$$

where for a partition $\mathcal{F}$ of S the matrix $A_{\mathcal{F}}$ is $|\mathcal{F}| \times m$ size matrix defined by

$$(A_{\mathcal{F}})_{T,j} = \pi_{\mathrm{sm},T}(g_j),$$

for $T \in \mathcal{F}$ and $j \in m$. Note that the matrix $A_{\mathcal{F}}$ has rows that access disjoint parts of the set-multilinear partition of $\bar{x}$, and so as the k-surjective permanent is set-multilinear with respect to rows, the computation $\mathrm{perm}_{|\mathcal{F}| \times m; k}(A_{\mathcal{F}})$ is set-multilinear with respect to the partition of $\bar{x}$. Further, a set-multilinear computation of this permanent will be set-multilinear with respect to $\bar{x}$. Hence, we can use the set-multilinear computation from Corollary 22 for the surjective permanent, which in this case yields a depth-4 set-multilinear formula of size $\mathrm{poly}(m, \Theta(\frac{|\mathcal{F}|}{\ln |\mathcal{F}|})^{|\mathcal{F}|})$. Computing $\pi_{\mathrm{sm},S}(g)$ is then a sum of $B_{|S|}$ many such $\mathrm{perm}_{|\mathcal{F}| \times m; k}(A_{\mathcal{F}})$ terms, which does not increase depth as we collapse two sequential layers of addition gates, and this preserves set-multilinearity. The resulting size of the expression is $\mathrm{poly}(m, \Theta(\frac{|\mathcal{F}|}{\ln |\mathcal{F}|})^{|\mathcal{F}|}, B_{|S|}, 2^d)) \leq \mathrm{poly}(m, \Theta(\frac{\ell}{\ln \ell})^{\ell}, 2^d)$, as $|S| = \ell$, $|\mathcal{F}| \leq |S|$. □

We now conclude with our set-multilinearization result over any field by gate-simulation.

Proof of Corollary 27. By gate simulation. Let Φ be the hypothesized circuit computing f. For each node v in Φ, split v into its at-most set-multilinear parts $\pi_{\mathrm{sm},S}(v)$ for each $S \subseteq [d]$. If $v = \alpha_1 v_1 + \cdots + \alpha_m v_m$, then we can express the at-most set-multilinear parts of v in terms of the v_i using a $\mathrm{poly}(m, 2^d)$-size product-depth 0 set-multilinear circuit by Lemma 25. If $v = v_1 \times \cdots \times v_m$, then we can express the at-most set-multilinear parts of v in terms of the v_i using a $\mathrm{poly}(m, \Theta(\frac{d}{\ln d})^d)$-size product-depth 2 set-multilinear circuit by Lemma 26. Correctness of the computation follows by induction on the circuit. The overall size of the circuit has increased from s to $\mathrm{poly}(s, \Theta(\frac{d}{\ln d})^d)$ by counting the size of the additional local gadgets of the gate simulation. Simulation of addition gates adds no product depth. Each product gate in the original circuit is turned into a product-depth 2 circuit in the gate simulation, hence the overall product-depth has at most doubled. □

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